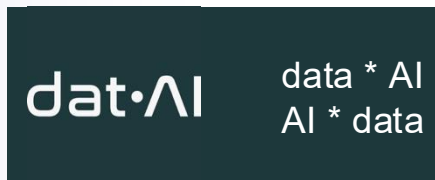


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LM3 *From Word Embeddings* *to Language Models*



From Word Embeddings to LLMs

Context and contextual word embeddings

Word2vec: Cons (1) - Word-level Embeddings

- ⌘ Word embeddings: one core meaning of a word (~ “weighted average” of its contexts)
- ⌘ Many words are polysemous, e.g., Paris, bush/Bush, play, etc.
- ⌘ In general, word meanings depends on the context
 - ⌘ “I **play** soccer”, “I **play** the violin” and “I went to see a **play** at the theater”.
 - ⌘ “**play**” => three different meanings, only one embedding in word2vec

Word2vec: Cons (2) – More Limitations

■ Rare / new words

- Empirical evidence suggest that the quality of the embeddings depends on frequency; what about rare and new words?
- Centroid of context words' vectors (baseline); better method by Lampinen & McClelland 2017 (<https://arxiv.org/abs/1710.10280>)

■ Tokens vs. phrases

- Word phrases referring to meaning units - especially names; e.g., New York, Wall street, etc.
- Learn word phrases first (Mikolov 2013, but see also [1,2], e.g., PMI - Pointwise Mutual Information, i.e., $\log P(x;y)/P(x)P(y)$)
- Compositionality (e.g., $\mathbf{v}(\text{"soccer team"}) = f(\mathbf{v}(\text{"soccer"}), \mathbf{v}(\text{"team"}))$) ... several papers, centroid as a baseline
- Disambiguate names or word phrases first (more details about this later, see below)

[1] <https://lilianweng.github.io/lil-log/2017/10/15/learning-word-embedding.html>

[2] <https://towardsdatascience.com/word2vec-for-phrases-learning-embeddings-for-more-than-one-word-727b6cf723cf>

Meaning in Context

bar

What does “bar” mean?

Meaning in Context

bar



noun [C]

UK /bɑːr/ US

bar

What does “bar” mean?

bar noun [C] (DRINKING)

Add to word list

A1
a place where alcoholic drinks are sold and drunk, or the area behind the person serving the drinks:

- I met him in a bar in Soho.

bar noun [C] (BLOCK)

B1
a small block of something solid:

- a chocolate bar

bar noun [C] (LONG PIECE)

B2

a long, thin piece of metal or wood:

- There were bars on the downstairs windows.

bar noun [C] (MUSIC)

one of the short, equal groups of notes that a piece of music is divided into:

- The band played the first few bars.

bar noun [C] (PREVENTING SUCCESS)

something that prevents you doing something or having something:

- Lack of money should not be a bar to a good education.

Polisemy:
many words have different meanings; we may call them *senses*

Meaning in Context

bar



noun [C]

UK /baːr/ US

(1) I (2) arrived (3) at (4) the (5) **bar** (6) tonight

What is the right meaning and which hints you use for the choice?

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▶ the bar

lawyers thought of as a group:

• Haughey was **called to the bar** (= became a lawyer) in 1949.

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Word meaning depends on the context

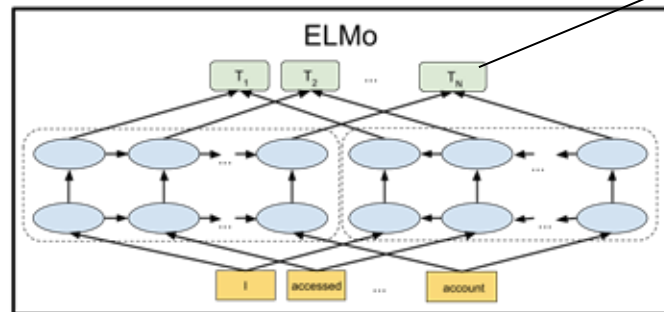
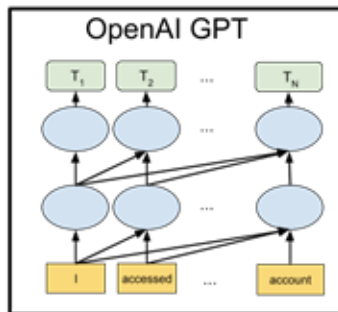
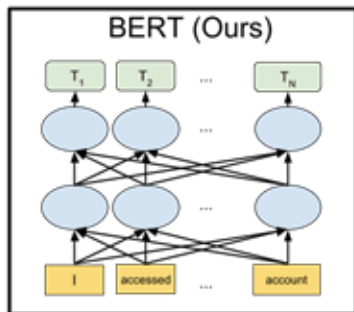
Contextual Word Embeddings

🔗 **Contextual word embeddings:** the embedding of each word depends on the context in which it appears, i.e., it is influenced by the embeddings of nearby words

🔗 **Different approaches:**

🔗 Transformers: BERT, OpenAI GPT (newer)

🔗 BI-LSTM: ELMo (2018)



Contextual embedded representations

Language Models vs Meaning in Context

- Language models (LMs)
 - Estimate probabilities of lexical units, e.g., words in a sentence
- LMs vs contextual word embeddings
 - Word representations optimized to compute the probabilities
 - Word representations will depend on the context the words appear in, i.e., on a specific occurrence of a word in a context

(1) I (2) stopped (3) at (4) the (5) **bar** (6) yesterday

Occurrence of “bar”

Meaning in Context



What is the right meaning and which hints you use for the choice?

(1) I (2) arrived (3) at (4) the (5) bar (6) tonight

bar noun [C] (DRINKING)

A1

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(1) I (2) arrived (3) at (4) the (5) bar (6) tonight

Contextualized word embedding intuition: the embedding depends on the context

Meaning in Context (Cons.)

(1) I (2) arrived (3) at (4) the (5) **bar** (6) tonight

(1) I (2) stopped (3) at (4) the (5) **bar** (6) yesterday

(1) I (2) drank (3) at (4) the (5) **bar** (6) with

.....

But then, each occurrence of bar ...

- (similar but) different context ...
- (similar but) different meaning ...

Contextualized word embedding intuition: the embedding depends on the context

The Semantic Granularity Spectrum

Synsets (wordnets) / entities

Word-level

Core meaning

- Word2vec
- GloVe

Sense-level

N-core meanings

- **Word2GM**

Sense-level meanings

- Li & Jurafsky (2015)
- ... (?)

Entity-level

- **TEE** (Bianchi & al. 2018)

Word + word phrases

Core meaning

- Annotated word2vec

Contextual word embeddings + language models

Occurrence-level

Sentence-specific meaning

- Elmo
- Bert
- PALM
- GPT-2
- GPT-3
- GPT-X...
- ...

n is fixed in advance for all the words



Can also provide embeddings of sentences, etc

Coarse grain

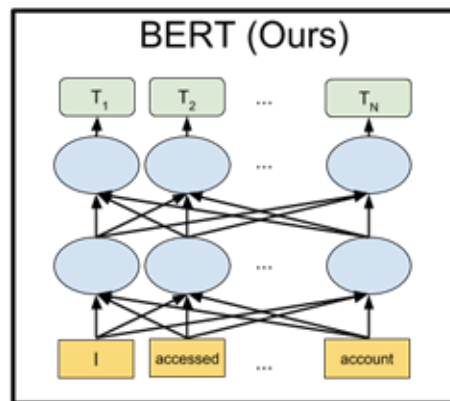
Example of application to learn entity embeddings

Fine grain

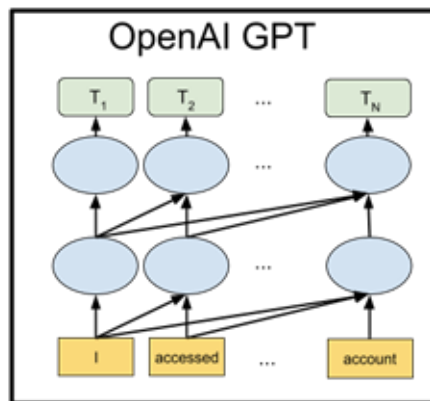
Key Features of Embedding-based (L)LMs

■ Two tasks

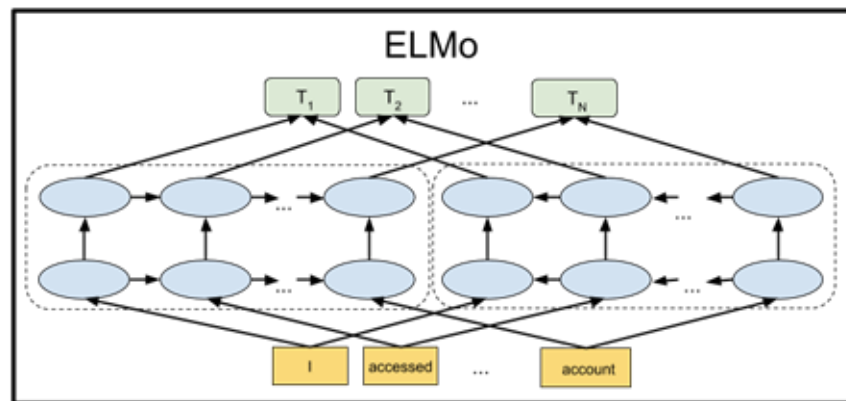
- Natural language understanding (NLU): feature extraction for downstream tasks, e.g., BERT
- Natural language generation (NLG), e.g., GPT



NLU



NLG

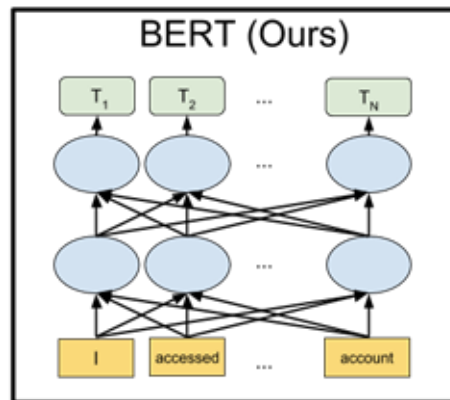


NLU

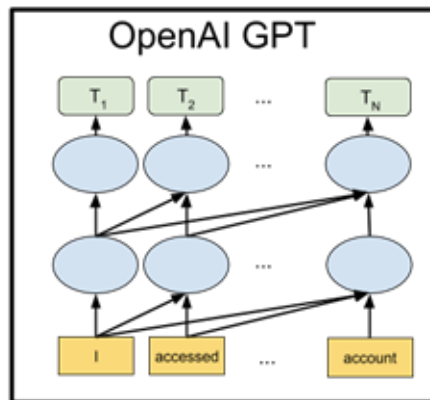
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NLU



NLG

Based on a powerful neural network architecture:
Transformers (!)

Next slides... intuition
Next presentation... details

Transformers: Transform Input Sequences

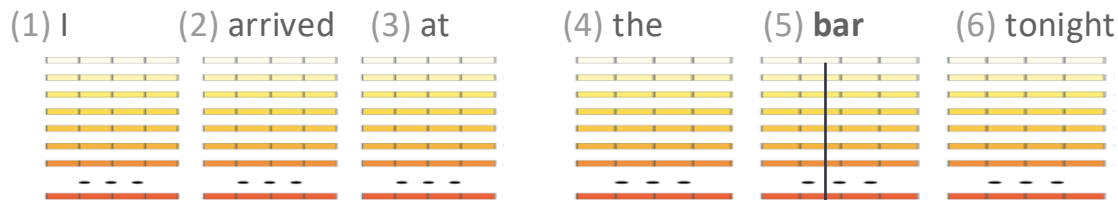
- Transforming an input sequence
 - Learn to transform an input sequence of **tokens**
 - ... into **embeddings**
 - ... into an output sequence of **tokens**
 - Principal mechanisms
 - transform the embeddings of the input tokens
 - multi-head **attention** (“attention is all you need”)
 - Different architectures combining at least one of two modules: encoders and decoders

Key Features of Embedding-based (L)LMs

■ Two tasks

- Natural language understanding (NLU): feature extraction for downstream tasks, e.g., BERT
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Meaning in Context with a Transformer Encoder

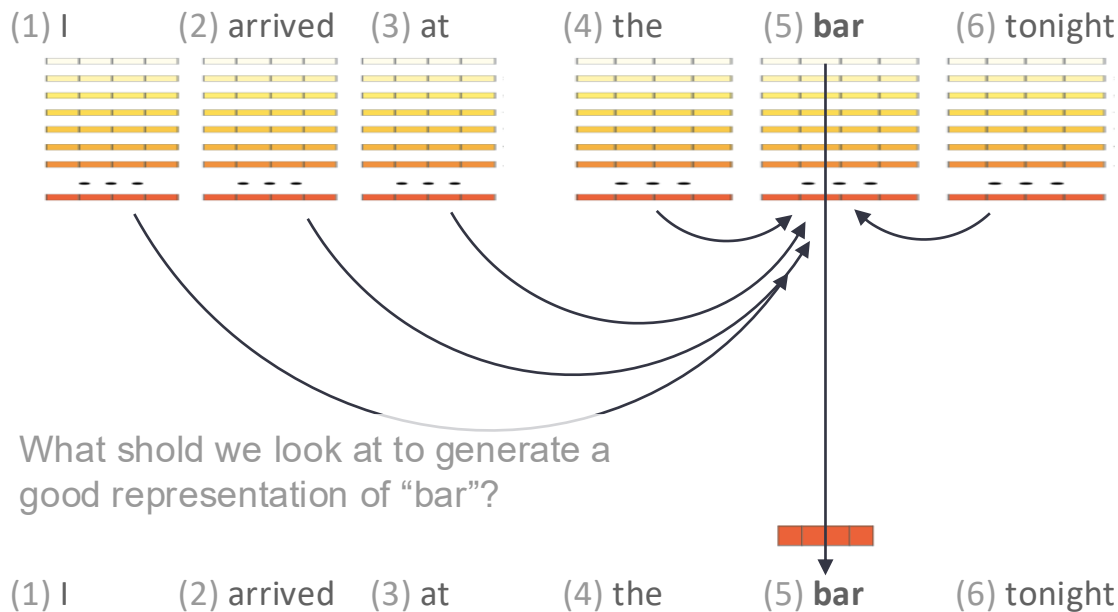


We are transforming input embeddings associated with input tokens

What should we look at to generate a good representation of “bar”?



Meaning in Context with a Transformer Encoder

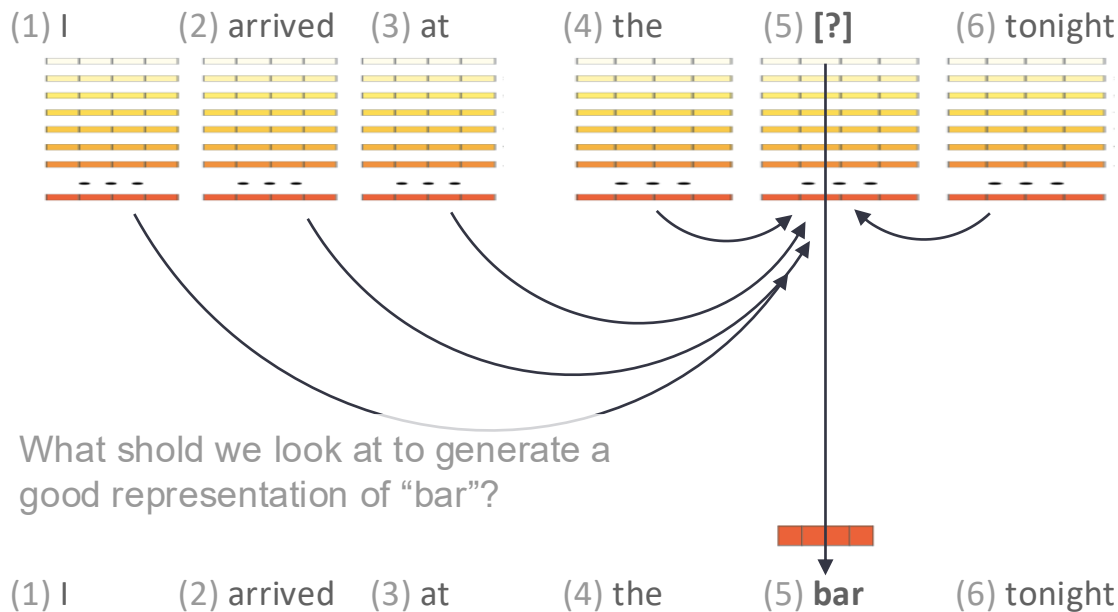


We are transforming input embeddings associated with input tokens

Idea: mix into its representation representations of other tokens

Context should help reconstructing the right meaning

Meaning in Context with a Transformer Encoder



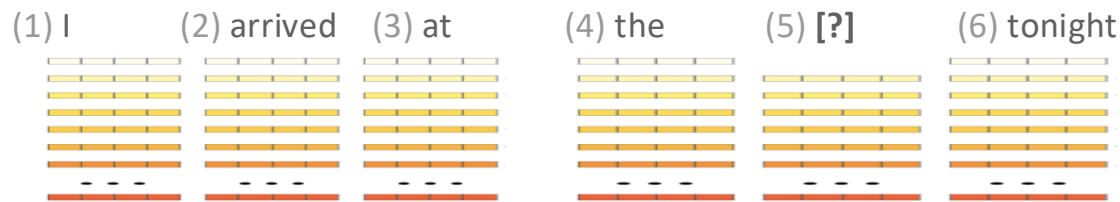
We are transforming input embeddings associated with input tokens

Idea: mix into its representation representations of other tokens

Masking: we train a model by masking a “word” and guess it by only looking at the context

Context should help reconstructing the right meaning

Meaning in Context with a Transformer Encoder



The generation of good representations of the input is referred to as **encoding**

Encoding (input)

We are interested in these vectors

(1) I (2) arrived (3) at (4) the (5) **bar** (6) tonight

Context should help reconstructing the right meaning

Context and Generation: the Extreme Case of Translation

(1) I (2) arrived (3) at (4) the (5) bar (6) tonight

← input
sequence

(1) [s] (2) je (3) suis (4) arrivè (5) au (6) bar (7) ce (7) soir

← output
sequence

Sequence to sequence (seq2seq)

Context and Generation: the Extreme Case of Translation

(1) I (2) arrived (3) at (4) the (5) **bar** (6) tonight

(1) [s] (2) je (3) suis (4) arrivè (5) au (6) **bar**

Sequence to sequence (seq2seq)

"Bar" (EN) translated in French

le bar	bar, bass, saloon, inn, cabaret, public house
la barre	bar, rod, helm, line, box, drawbar
le barreau	bar, rung, railing, well
la barrette	bar, barrette, biretta, cap, slide, hairclip
la mesure	measure, step, metering, meter, bar, moderation
le comptoir	counter, bar, post
le café	cafe, coffee, bar, java
l' obstacle	obstacle, barrier, impediment, hurdle, hindrance, bar
la bande	band, strip, tape, stripe, gang, bar

input
sequence

output
sequence

+

Context and Generation: the Extreme Case of Translation

(1) I (2) arrived (3) at (4) the (5) bar (6) tonight

← input
sequence

(1) [s] (2) je (3) suis (4) [?]

← output
sequence

What should we look at to predict “arrivé”?

Sequence to sequence (seq2seq)

Context and Generation: the Extreme Case of Translation

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← input
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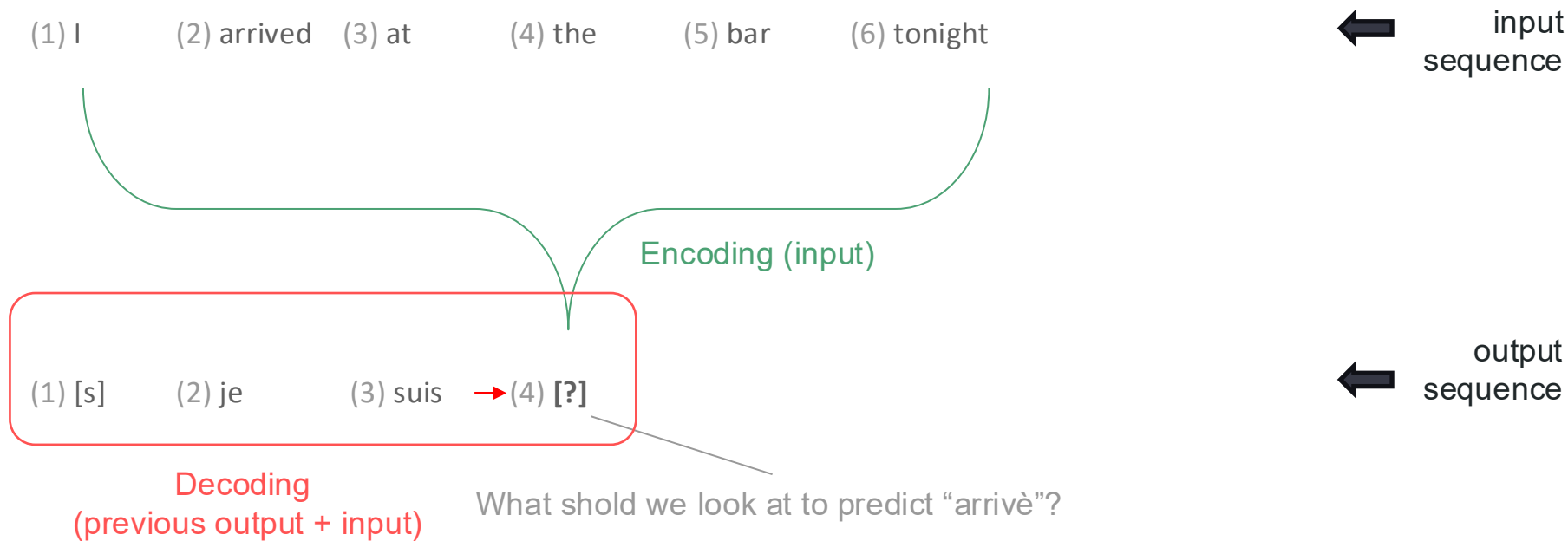
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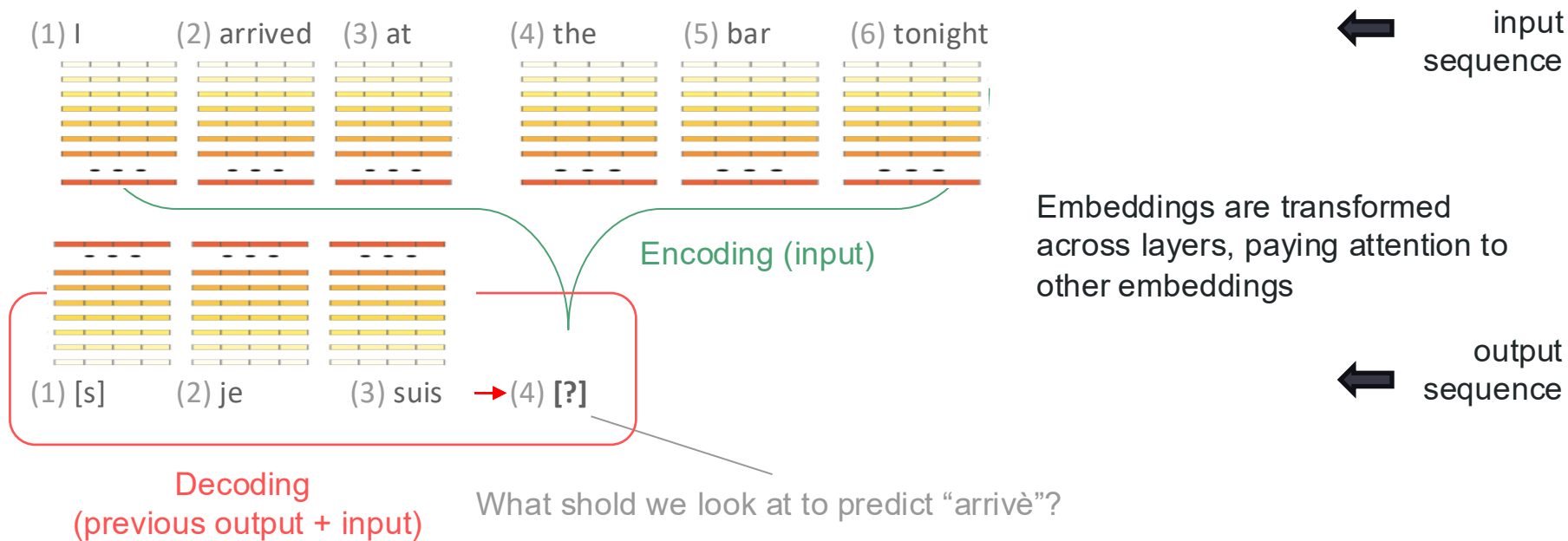
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Context and Generation: the Extreme Case of Translation



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Context and Generation: the Extreme Case of Translation



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Context and Generation:

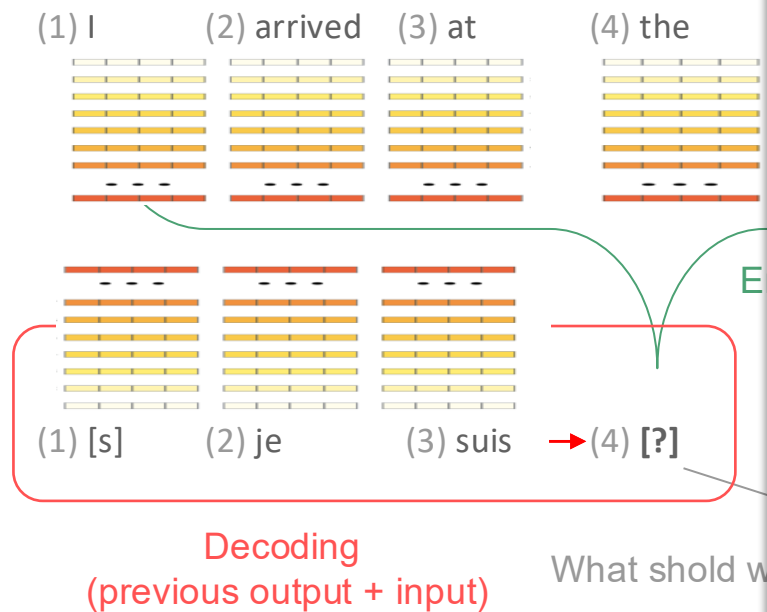


Figure 1: Applying the Transformer to machine translation. Source: [Google AI Blog](#).

Key Features of Embedding-based (L)LMs

■ Two tasks

- Natural language understanding (NLU): feature extraction for downstream tasks, e.g., BERT
- Natural language generation (NLG), e.g., GPT

■ Self-supervised pre-training

- Resource-intensive
- Data: large / extra-large corpus
- Tasks: mostly based on language modeling
- Type: self-supervised

■ Fine-tuning on specific data

- Less resource-intensive
- Different approaches
 - NLU
 - Downstream task, e.g., text classification
 - NLG
 - Better generation
 - Task-specific generation

Transformers: some references

- Detailed presentation of transformers:
 - [Jurafsky's book](#)
 - <https://jalammar.github.io/illustrated-transformer/> (IMHO best end-to-end explanation of transformers)
 - Lucas Beyer talk at M2L Mediterranean Machine Learning Summer School: <https://www.m2lschool.org/talks>
- Detailed presentation of BERT
 - <http://jalammar.github.io/illustrated-bert/>